CELIA RUBIO-MADRIGAL

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I work on deep learning for graph data: on how input structures shape data representation and computation, on generalization trade-offs, and on architectural efficiency. I am a la Caixa Fellow, a x2 HLF attendee, and EGMO IT.

Experience

- **Doctoral researcher** – *CISPA Helmholtz Center for Information Security* **2023-now**
  – Graph ML, efficiency, benchmarking, rewiring. – Advisor: Dr. Rebekka Burkholz
- **RA: Departmental collaboration** – *Ministerio de Educación & Universidad Complutense de Madrid* **2021-22**
  – Identifying complex Boolean functions with ML – Advisor: Dr. Ismael Rodríguez Laguna
- **REU: Introduction to research ICMAT-Severo Ochoa** – *Instituto de Ciencias Matemáticas (ICMAT)* **2021**
  – Differential geometry methods in ML applications – Advisor: Dr. David Martín de Diego
- **Intern: Consultant Developer** – *Guidewire Software* – NLP search engine to find internal raised issues. **2021**

Education

- **PhD in Computer Science** – *CISPA & Universität des Saarlandes (Saarbrücken, Germany)* **2023-now**
- **MSc in Advanced CS with AI** – *University of Strathclyde (Glasgow, UK)* – 8.57/10 (Distinction) **2022-23**
  – Recipient of the "la Caixa" Fellowship – Department's Best Overall Performance Prize (Top 1)
- **BSc in Mathematics** – *Universidad Complutense de Madrid (Spain)* – 9.08/10 **2017-22**
- **BSc in Computer Science** – *Universidad Complutense de Madrid (Spain)* – 9.16/10 **2017-22**

Publications

Legend: **bold**: First-author A* conference. * Equal contribution. † Co-supervision.

R-M., Burkholz. *Linear GCNs Need Better Bias, Not More Expressive Power.* GFM Workshop @ ICML'26.

Joshi, Gadhikar†, R-M.†, Dutta, Singh, Burkholz. *Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning.* MemFM Workshop @ ICML'26.

R-M., Burkholz. *Fixed Aggregation Features Can Rival GNNs.* **ICML'26**.

Reddy, Verma, R-M., Muandet, Burkholz. *Boosting for Predictive Sufficiency.* ICLR'26.

Reddy, R-M., Burkholz, Muandet. *When Shift Happens - Confounding Is to Blame.* ICLR'26.

Jacobs, Gadhikar, R-M., Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity.* ICLR'26.

R-M.*, Jamadandi*, Burkholz. *GNNs Getting ComFy: Community and Feature Similarity Guided Rewiring.* **ICLR'25**.

Jamadandi*, R-M.*, Burkholz. *Spectral Graph Pruning against Over-squashing and Over-smoothing.* **NeurIPS'24**.

Awards

– 13th Heidelberg Laureate Forum: *200 international young researchers network with math and CS laureates* **2026**
– Dept. Best Overall Performance Prize: *University of Strathclyde's top student in MSc Advanced CS* **2023**
– Nova 111 Student List: *Spain's top 10 talents under 25 in Computer Science* **2023**

– 9th Heidelberg Laureate Forum. Featured in: *HLFF Spotlight*, *Tribuna Complutense* **2022**
– "la Caixa" Fellowship: *Full funding for MSc studies.* Featured in: *El País*, *El Plural*, *TeleMadrid* **2022**

- **Undergraduate**

– Aula de Verano "Blas Cabrera" for Spain's top undergrad records: *Intl. Univ. Menéndez Pelayo (UIMP)* **2022**
– SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* **2022**
– Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* **2018-22**

- **Pre-undergraduate**

– Premio Extraordinario de Bachillerato: *Madrid's top 25 in general-content exam (1000€)* **2017**
– EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* **2017**

Talks

- **Invited talks** – *Seminars and reading groups* **2025-26**

– GLOW, TUWien ML RU seminar, Cohere For AI, CISPA ML seminar, AI Security Afternoon.

- **Contributed talks** – *Conferences and workshops* **2024-26**

– GSP 2026, HAICON 2024. Lightning talks: GTML 2025, MLG 2025 (2), NetSci 2025.

- **Non-archival posters** – *Non-archival conferences, workshops, and sessions* **2024-26**

– GSP 2026, CPAL 2026 (3), WiML 2024 2025, ELLIS Pre-NeurIPS/ICML 2024 2025 2026.

Teaching

- **Math Olympiad Trainer** – *International EGMO Preparatory Seminar in Bonn, Germany* **2026**
- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes (3)* **2024-now**
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – *University of Strathclyde* **2023**

– "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel

Service

- **Academic service** **2024-now**

– Reviewer: MLG 2025-26, ICLR 2026 – **Top 200 (1%)**, CPAL 2026, ICML 2026 – Gold (25%), NeurIPS 2026.
– Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.

- **Math Olympiad service**

– IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers **2023-now**
– Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. **2021-22**
– Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. **2020-24**

- **Student representation** – *Facultad de Informática (FdI), Universidad Complutense de Madrid* **2020-22**

– Secretary of the FdI Student Council 2021-22. Member of the FdI Faculty Board 2020-22.

- **LaTeX package development**

– **`mousik`**: Music rendering made simple for small excerpts with LaTeX. **2026**
– **`ddphonism`**: Twelve-tone music notation diagrams. Included in MiKTeX and TeXLive. **2019**

- **Mentoring and outreach**

– Mentor for new "la Caixa" Fellows **2023-25**
– Outreach articles for DivulgaMAT (RSME journal): *Serialismo y matemáticas* **2019**
– Outreach videos for the Spanish math education community: **2018-21**